FRD500 Quick Manual (User's Manual)

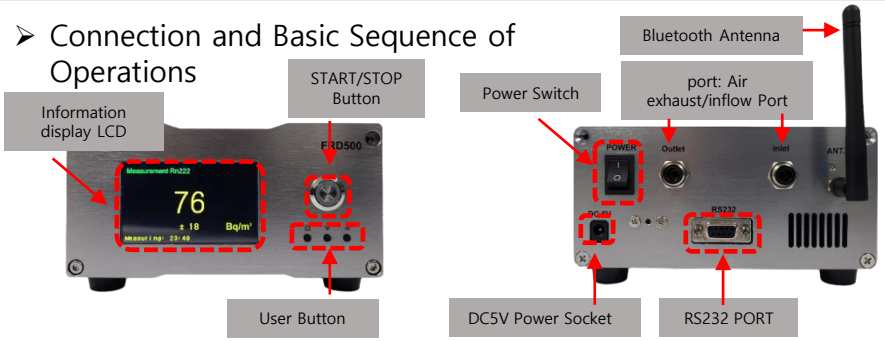


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※ This device is an electromagnetic compatible device for business use(Class A)that the seller or the user should be aware of, and is intended for use besides home.

FRD500 Quick Manual

➢ Connection and Basic Sequence of Operations



1. Connect the Antenna to the Antenna Port on the back of the FRD500.
2. Connect the supplied dust filter to the Inlet to prevent the inflow of dust.
3. Connect the DC 5V/2A adaptor provided with the DC 5V power connector.
4. When the Power switch is turned on, the screen on the LCD display in the front turns on and starts to operate.
5. Before pressing Start Measurement, connect the device to the app to synchronize the current time with the internal clock.
6. Use ∨ Button to select the measurement mode in the Measurement Mode Selection screen.
7. Press the Start/Stop button to start the measurement with the LED light on. (Data storage is also performed at one hour intervals.)
8. When completing the measurement, press Start/Stop button to finish the measurement. (Data is automatically saved when measured for more than 1 hour.)
9. You can check and save on the app or PC program during measurement. (Real-time measurement is displayed when set at Auto in PC program)
10. Using a pump causes pressure difference inside the device. Therefore, it is impossible to operate the button for 3 seconds to have the internal pressure equilibrium at the end of the measurement.

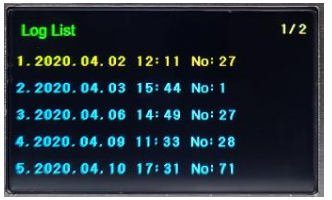
➢ Screen configuration, Button operation



① Displays the mode currently being measured
② Displays Radon measurement result and erro
③ Selects display unit (Bq/m3 or pCi/L)
④ Status Indication
Process : Total measurement time
Measuring: Time remaining before the measured value is updated
oC, % hPa : Temperature, Humidity, Barometric pressure
Peak: Displays Peak Value during the measurement period
⑤ START/STOP : LED flashes when started with ON/OFF Button
⑥ BUTTON □ : Switches the screens of Main, Log List, and Information
BUTTON ∨ : Main Screen - Changes Measurement mode
Log List Screen - Changes Log Data
BUTTON ◯ : Main Screen - Selects Measurement mode
Log List Screen - Selects Log Data
Log Data Screen – Turns over the page

➢ Log List Screen

① Pressing □ Button on the measurement screen changes to the Log List View screen.
② Log List shows the year, month, day, and time on which the first data was stored since the start of measurement, and the number of stored data is displayed. (The FRD500's internal clock is automatically updated to the current time when accessing the app..)
③ The selected Log List is expressed in yellow. (To change selection, use ∨ button, and to check the data of the selected Log list, use ◯ button..)
④ The current page and the maximum page of the list are shown in the upper right corner.(Up to 5 lists on one page)
⑤ It consists of up to 10 slots and can store data up to 60 days at one hour interval per slot.
⑥ When one slot is full, the data is automatically saved to the next slot, and when all 10 slots are used, the oldest Slot 1 is automatically erased. Slots 2 to 10 are rearranged to Slots 1 to 9, and data is stored in Slot 10.



➢ Review Screen

① Displays 12 measurements in one screen.
② The information displayed on the bottom of the screen shows that the year, month, day and hour of the first data was stored and the number of stored data, and the measurement mode is shown in the parentheses.
③ The entire pages and the current page are shown in the upper right corner.
(Use ◯ Button to flip the page forward,
Use ∨ Button to flip the page backward..)
④ Use □ Button to return to the Main screen.



➢ FRD500 Specifications

- Type : pulsed ion chamber 400cc
- Sensitivity : 0.9cpm/pCi/L at 10pCi/L (54cph/pCi/L)
- Range Air (Rn222) : 1 ~ 111,000 Bq/m³ (0.1 ~ 3,000 pCi/L)
  Air (Rn222+Rn220) : 1 ~ 111,000 Bq/m³ (0.1 ~ 3,000 pCi/L)
- Error :< ±5% at 10pCi/L (min. error < ±0.4pCi/L)
- Reproducibility :< ±5% at 10pCi/L
- Data Saving Interval : 1h
- Operating Range : 0 ~ 50℃, RH < 90%
- Internal Sensor : Temp., Humidity, Barometer
- Air Supply : Micro Air Pump (1 LPM)
- Power : DC 5V, 1.4A (using 5V, 2A DC adapter)
  * FRD500-only external battery pack uses a 12V adaptor, Be careful not to use mixed with other type packs
- Size : W160 x D230 x H90(mm)
- Data Communication : Bluetooth LE, RS232
- Data Log : Total 14,400 points (60 days X 10 slots)
- Display : 3.2 inch TFT-LCD
- Weight : 1.6 kg

